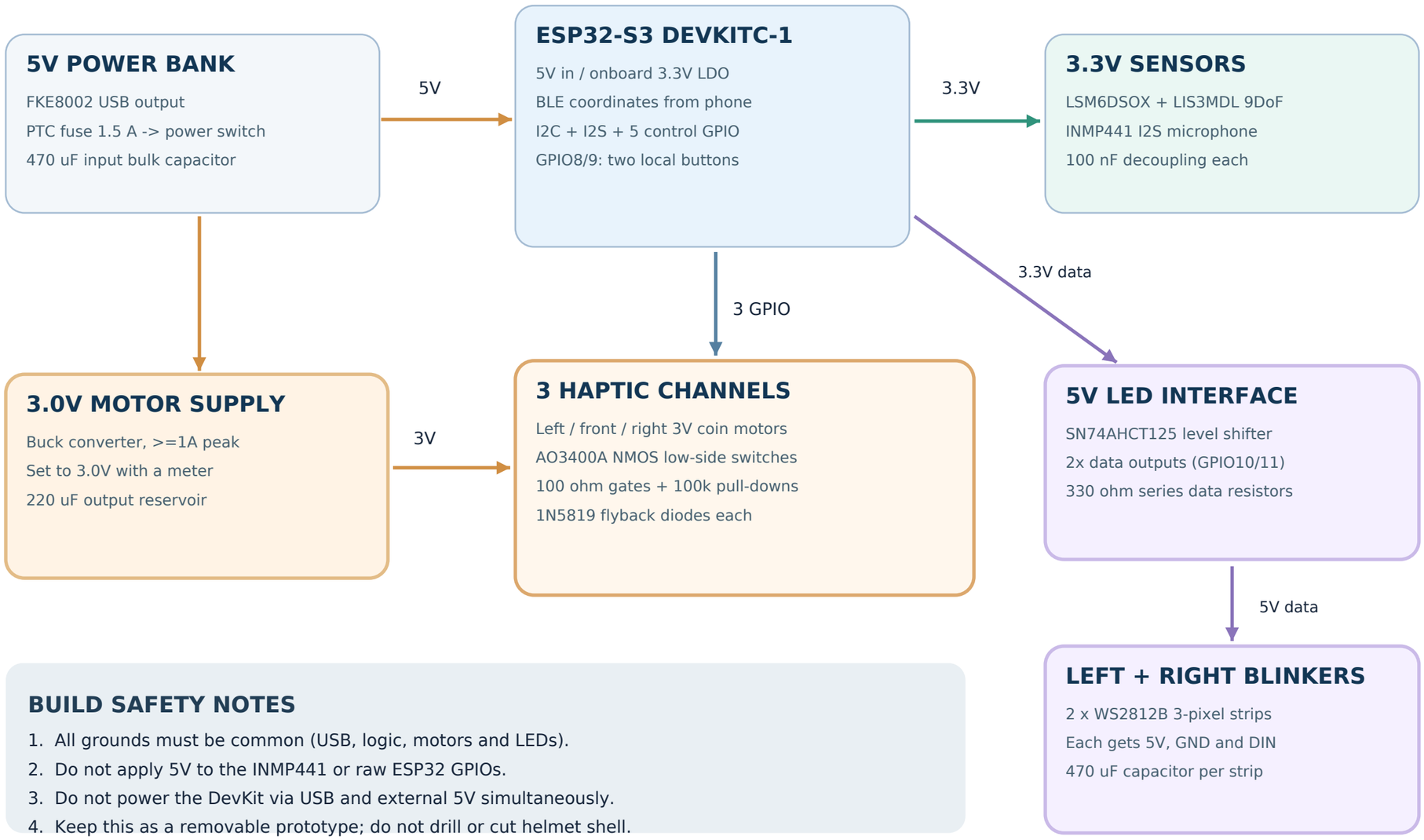
SMART HELMET / ELECTRICAL WIRING PLAN

Prototype revision A | Smartphone GPS over BLE | Regulated USB power bank



BUILD SAFETY NOTES

1. All grounds must be common (USB, logic, motors and LEDs).
2. Do not apply 5V to the INMP441 or raw ESP32 GPIOs.
3. Do not power the DevKit via USB and external 5V simultaneously.
4. Keep this as a removable prototype; do not drill or cut helmet shell.

KiCad source: Helmet_Navigation.kicad_sch | View net labels to trace every connection.

PIN MAP & ASSEMBLY CHECKS

Net / pin	Destination	Critical detail
GPIO4 / GPIO5	IMU SDA / SCL	3.3V I2C bus
GPIO6	IMU INT1	Optional motion interrupt
GPIO8 / GPIO9	MODE / SOS buttons	Use internal pull-ups
GPIO10 / GPIO11	Left / right LED data	Via SN74AHCT125 at 5V
GPIO14 / 15 / 16	I2S BCLK / WS / SD	Mic VDD is 3.3V
GPIO17 / 18 / 21	Left / front / right motors	Through AO3400A gates
+5V_SYS	ESP32 5V, LEDs, AHCT125, buck	USB bank via switch/fuse
+3V3_LOGIC	IMU + microphone	DevKit regulated output
+3V0_MOT	Three coin motors	DC/DC buck output

BEFORE APPLYING POWER

[ ] Buck output measures 3.0V, unloaded and loaded.

[ ] MOSFET pinout and flyback diode polarities match part markings.

[ ] 5V strip has common GND and receives DATA only through 5V buffer.

[ ] INMP441 and IMU see 3.3V; no 5V at signal pins.

[ ] Run MCU, then sensors, then one motor, then LEDs, then all loads.

Reference only: prototype module wiring; not a PCB, production safety certification or validated helmet system.